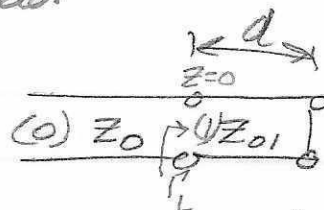


ECE 635 Homework #5 Answers

1. 23.1 W at the input and load

2. In the figure I labeled



so in region (1)

$$V_1 = \frac{-V_0(1+\Gamma_L)}{\sin(k_1 d)} \sin[k_1(z-d)], \quad \Gamma_L = \frac{jZ_{01} \tan(k_1 d) - Z_0}{jZ_0 \tan(k_1 d) + Z_{01}} \quad (7)$$

Solution $\rightarrow$ and $E_1 = \frac{-E_0(1+\Gamma_L)}{\sin(k_1 d)} \sin[k_1(z-d)], \quad \Gamma_L = \frac{j\eta_{01} \tan(k_1 d) - \eta_0}{j\eta_0 \tan(k_1 d) + \eta_{01}} \quad (8)$
for figure

also carrying (8) out further

$$1 + \Gamma_L = \frac{j\eta_{01} \tan(k_1 d) + \eta_0 + j\eta_{01} \tan(k_1 d) - \eta_0}{j\eta_0 \tan(k_1 d) + \eta_{01}}$$

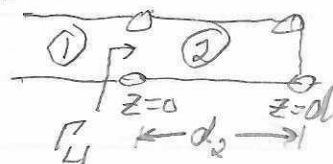
$$= \frac{2j\eta_{01} \tan(k_1 d)}{j\eta_0 \tan(k_1 d) + \eta_{01}} \quad (9)$$

when (9) is substituted into (8)

$$E_1 = \frac{-E_0}{\sin(k_1 d)} \frac{2j\eta_{01}}{[j\eta_0 \tan(k_1 d) + \eta_{01}]} \frac{\sin(k_1 d)}{\cos(k_1 d)} \sin[k_1(z-d)]$$

alternate solution $\rightarrow E_1 = \frac{-2jE_0\eta_{01} \sin[k_1(z-d)]}{\cos(k_1 d) [j\eta_0 \tan(k_1 d) + \eta_{01}]} \quad \text{alternate (8)}$

If you used the notation



$$V_2 = \frac{V_{01}(1+\Gamma_{L1})}{\sin(k_2 d_2)} \sin[k_2(z-d_2)]; \quad \Gamma_{L1} = \frac{jZ_{02} \tan(k_2 d_2) - Z_{01}}{jZ_{01} \tan(k_2 d_2) + Z_{02}}; \quad d_2 = d$$

$$E_2 = \frac{-E_{01}(1+\Gamma_{L1})}{\sin(k_2 d_2)} \sin[k_2(z-d_2)]; \quad \Gamma_{L1} = \frac{j\eta_{02} \tan(k_2 d_2) - \eta_{01}}{j\eta_{01} \tan(k_2 d_2) + \eta_{02}}$$

Or $E_2 = \frac{-2jE_0\eta_{02} \sin[k_2(z-d_2)]}{\cos(k_2 d_2) [j\eta_{02} \tan(k_2 d_2) + \eta_{01}]}$

3.

$$V_1 = V_{01} e^{-jk_1 z} ; V_2 = V_{01} / (1 - \Gamma_{L2}) e^{j2k_2(z-d)}$$

$$V_3 = -V_{01} \sqrt{\frac{Z_{03}}{Z_{01}}} e^{-jk_3(z-d)} \quad \text{K eqn. 11)$$

dual

$$E_1 = E_{01} e^{-jk_1 z} ; E_2 = E_{01} / (1 + \Gamma_{L2}) e^{j2k_2(z-d)}$$

$$E_3 = -E_{01} \sqrt{\frac{\eta_{03}}{\eta_{01}}} e^{-jk_3(z-d)} ; \Gamma_{L2} = \frac{\eta_{03} - \eta_{02}}{\eta_{03} + \eta_{02}} \quad \text{K eqn. 11)$$

$$P_3 = \frac{1}{2} \operatorname{Re} \left\{ \frac{|V_{03}|^2}{Z_{03}} \right\} = \frac{|V_{03}|^2}{2Z_{03}} = \frac{V_{01}^2}{2} \left(\frac{Z_{03}}{Z_{01}} \right) \frac{1}{Z_{03}} = \frac{V_{01}^2}{2Z_{01}}$$

$$P_1 = \frac{1}{2} \operatorname{Re} \left\{ \frac{|V_{01}|^2}{Z_{01}} \right\} = \frac{1}{2} \frac{|V_{01}|^2}{Z_{01}}$$

dual

$$P_3 = \frac{1}{2} \operatorname{Re} \left\{ \frac{|E_{03}|^2}{\eta_{03}} \right\} = \frac{E_{01}^2}{2\eta_{01}}$$

$$P_1 = \frac{1}{2} \operatorname{Re} \left\{ \frac{|E_{01}|^2}{\eta_{01}} \right\} = \frac{E_{01}^2}{2\eta_{01}}$$